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Interpretable machine learning with SHAP for esports performance analysis of professional counter-strike players from 2012 to 2025

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Abstract

Performance analysis in traditional sports has become a mature area of sports science, whereas its application in esports remains relatively limited. This study applies an interpretable machine learning framework combining SHAP analysis and five classic algorithms to evaluate the performance of professional Counter-Strike players. Among the models tested via five-fold cross-validation, XGBoost demonstrated the best performance. The dataset comprises 1468 professional players from the Counter-Strike: Global Offensive period (2012–2023) and 880 from the Counter-Strike 2 period (2023–2025), with each player evaluated using 39 advanced in-game behavioral metrics categorized into seven skill dimensions (e.g., firepower, entrying, trading, clutching, opening, sniping, utility). Results show that key performance indicators such as Kills Per Round (KPR), Opening Success (OS), Rounds With a Kill (RWK), Rounds With a Multi-Kill (RWMK), Support Rounds (SR), Pistol Round Rating (PRR), and Saves Per Round Loss (SPRL) consistently ranked among the top ten features in both periods. The top six influential features of the model reveal interactions between features and exhibit a positive correlation with SHAP values. The case diagnostics further enabled precise evaluation of individual strengths and weaknesses. This exploratory study provides empirical insights and an interpretable machine learning analytical paradigm for performance evaluation and talent identification in esports.

Keywords

Game behaviour metrics, tactical skill, talent development

Introduction

Esports is a type of competitive sports activity conducted on electronic platforms, typically computers, consoles, or mobile devices, where individuals or teams compete against each other in structured game competitions.[1,2] Esports exhibit similar core attributes of traditional sports, including competitiveness, spectator engagement, standardized rules, and organized structures.[3,4] In recent years, esports has experienced rapid growth, attracting worldwide attention and generating significant economic value.[5] The domain encompasses a wide range of game genres, such as First-Person Shooters (FPS), Multiplayer Online Battle Arenas (MOBA), Real-Time Strategy (RTS), and sports simulation games.[1] Although various factors have contributed to ongoing controversies and skepticism surrounding esports,[6] it is undeniable that the field is progressing steadily toward professionalization and sportization.[7] Among esports games, Counter-Strike (CS) stands out as a pioneering and enduring FPS franchise. With decades of development, Counter-Strike has remained one of the most popular and economically impactful esports games. Clearly, the explosive growth of esports, exemplified by games like Counter-Strike, represents a new paradigm of sports in the digital age, transcending pure competition to become a cultural phenomenon and an emerging professional career path.[8–10] Although previous studies have shown that the definition of esports encompasses multiple dimensions, organization and competitiveness are regarded as essential elements of esports.[11] With this background, we recognize

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that esports, like traditional sports, generates standardized competition data and player performance metrics suitable for quantitative evaluation. However, there is still very little research applying interpretable machine learning to evaluate esports players' performance compared to traditional sports, such as table tennis and football.[12–14] The existing research on Counter-Strike esports primarily focuses on match outcomes or economic decision-making, rather than players' performance evaluation.[15]

In traditional sports, well-established performance analysis systems and methodologies have been developed to evaluate matches data and assess talent.[16,17] In contrast, esports has not yet been fully integrated into the domain of sports science, and its training and performance evaluation methods remain in a developmental stage.[7] Although esports, particularly First-Person Shooter (FPS) games are becoming increasingly professionalized, the evaluating framework of players' performance are still relatively underdeveloped.[5] However, studies have shown that the fundamental elements of esports, such as players, teams, and competition formats are highly analogous to those found in traditional sports.[18] Esports and traditional sports also share commonalities in capability structures and training frameworks, including dimensions such as reaction time, psychological resilience, and physical training.[5,19] In both esports and traditional sports, the ultimate goal is to maintain a high level of performance and secure victory through rigorous skill development in high-pressure competitive environments.[5,20] Moreover, Counter-Strike features highly professional and structured player action data,[21] which provides a solid foundation for conducting performance analysis. These similarities and compatibilities not only help in understanding the essence of esports as a reasonable sports activity, but also support the application of traditional sports performance analysis paradigms to the context of Counter-Strike player evaluation. This alignment establishes a strong conceptual basis for the present study's adaptation of traditional performance analysis frameworks to the esports domain.

Counter-Strike 2 (CS2) is a prototypical tactical first-person shooter game in which two five-player teams compete in timed rounds, with one side playing as Terrorists (T) and the other as Counter-Terrorists (CT). The T side aims to plant and detonate a bomb at one of two designated sites, while the CT side must prevent the bomb from being planted or defuse it before detonation.[22] Moreover, eliminating all opponents within the specified time limit can also win a round. This form of esports has structural similarities to traditional team sports in terms of strategic competition and team coordination.[23,24] Each professional game generates rich embedded behavioral data from players, which is essentially a record of structured data in game logs transmitted between the client and the game server,[15,24] such as damage, utility usage efficiency, and shooting accuracy. HLTV is the most reputable platform for CS-related data, news, and match reports, offers extensive open-access datasets on player and match performance,[21] functionally similar to football data platforms such as Transfermarkt. However, the evaluation of CS players still largely relies on superficial statistical comparisons, and the identification and selection of top-performing players often lack scientifically grounded systems. This gap limits the esports industry's capacity to accurately identify high-potential talent and delays its progression toward a more structured and scientific sporting model. Therefore, this study integrates interpretable machine learning (ML) techniques and SHAP (SHapley Additive exPlanations) framework into CS performance analysis. In the traditional sports domain, interpretable ML models have been successfully applied to match performance analysis and player evaluation.[12,14] This provides us with methodological references and examples of how to conduct such research in esports.

Although there are some researches have begun to apply machine learning techniques to predict match outcomes or analyze in-game economic behaviors in Counter-Strike,[25] studies specifically focused on the performance of professional CS players remain limited. Moreover, considering the version changes and tactical evolution, this study adopts a longitudinal perspective by incorporating data from CS: GO (2012–2023) and CS2 (2023–2025). By analyzing player performance across the CS:GO (2012–2023) and CS2 (2023–2025) periods, our study offers a more comprehensive and nuanced view of how key performance indicators evolve between these periods.

In summary, in the background of Counter-Strike esports, comparison of multiple machine learning models, and an interpretable SHAP-based analytical framework, this study primarily aims to: (1) conduct a global interpretation of key performance features that distinguish professional players across two competitive eras, Counter-Strike: Global Offensive (2012–2023) and Counter-Strike 2 (2023–2025); (2) explore individual players' strengths and weaknesses at the micro level through case-based diagnostic analysis. These objectives are based on the assumptions that (a) esports player performance can be effectively modeled using machine learning algorithms; (b) player performance feature importance varies across different periods; (c) there may be some interaction between player performance metrics; and (d) there may be a linear or nonlinear relationship between player metric importance value and the metrics' original value. These underlying objectives and assumptions guided the analytic approach and ensured the relevance of the findings to performance assessment, talent identification, and training in the evolving field of esports. By exploring the longitudinal perspective, this study aims to identify how the importance of key gameplay features has evolved alongside version updates and player development.

Methods

Research design

This study employed a machine learning framework combined with SHAP-based interpretability analysis to

investigate the performance of professional players in the Counter-Strike (CS) esports domain across two critical stages, including 1468 pro players in the CSGO (2012–2023) period and 880 pro players in the CS2 (2023–2025) period. The dataset was sourced from HLTV (https://www.hltv.org/), a globally recognized platform dedicated to competitive Counter-Strike, offering news, match schedules, live scores, player statistics, photographs, and a vibrant community forum.

Several mainstream machine learning models were evaluated, including XGBoost (XGB), LightGBM, Random Forest (RF), Decision Tree (DT), and Logistic Regression (LG). Their performance was assessed using metrics such as AUC, Accuracy, Precision, F1-score, and Recall. XGBoost outperformed other models in both CS:GO (2012–2023) and CS2 (2023–2025) periods of five-fold cross-validation and was therefore selected as the primary model for subsequent SHAP analysis. All data preprocessing, model training, visualization, and SHAP analysis were conducted in Python. The overall research design is illustrated in Figure 1.

Data resources and selection

This study focuses on the development trajectory of the Counter-Strike: Global Offensive (CSGO) esports title, which spans two main phases: the CSGO period (2012–2023) and the CS2 period (2023–2025). The research team collected performance data for 1468 professional players during the CSGO phase and 880 players during the CS2 phase. Each player's performance is described using 39 game behavioral indicators categorized under seven tactical skill dimensions (e.g., firepower, entrying, trading, clutching, opening, sniping, utility) as defined by HLTV's Advanced Stats system.

The data were obtained on June 12, 2025, from Half-Life Television (https://www.hltv.org/), the most authoritative statistical and analytical platform in the CS esports ecosystem. The data for this study was manually collected by the research team from the HLTV official website and is only used for academic analysis purposes. HLTV's data has been widely used and validated in previous academic studies.[21,26,27]

The target variable (Y) in this study is the players' Rating, a widely accepted composite metric used to assess individual performance in CS.[24] Compared to single-dimensional metrics such as kill-death ratio (KDR), average damage per round (ADR), or KAST%, the rating provides a more holistic evaluation of player impact. Prior studies have recognized the rating as a robust indicator of competitive value.[21] For each game version, although the HLTV rating formula differs slightly, the threshold bands are identical. Based on HLTV's standardized scale, players with a rating ≥ 1.05 are categorized as Good, and those with a rating between 0.95 and 1.04 as Okay. Accordingly, we construct a binary label by mapping Good (≥ 1.05) to $y = 1$ and Okay (0.95–1.04) to $y = 0$. All preprocessing, model training, and SHAP analyses are conducted separately within each period. During the CS:GO period, the number of players with label 1 was 453, accounting for approximately 30.9% of the total. During the CS2 period, the number of players with label 1 was 584, accounting for approximately 66.3% of the total.

During the CSGO period, HLTV Rating 1.0 was used for ranking, while Rating 2.0 has been adopted for the CS2 era. The Rating 1.0 formula is as follows:

$$Rating_{1.0} = \frac{Rating_K + 0.7 \times Rating_S + Rating_{MK}}{2.7}$$

Where $Rating_K$, $Rating_S$, $Rating_{MK}$ represent kill, survival, and multi-kill components, respectively. The formula emphasizes the higher contribution of kills relative to survival.[24] Although the Rating 1.0 algorithm is publicly available, the complete method behind Rating 2.0 remains undisclosed. Rating 2.0 incorporates additional metrics such as KAST (Kill/Assist/Survival/Traded), ADR (Average Damage per Round), and contextual adjustments for T and CT sides.

Rating was selected as the target variable for three main reasons: (1) it is a comprehensive reflection of both offensive and tactical contributions in gameplay; (2) it is widely used in professional scouting, ranking systems, and commentary, making it a practical and industry-recognized benchmark; and (3) it provides a standardized, continuous score suitable for interpretable classification modeling. By focusing on the distinction between "good" ($\geq$1.05) and "okay" (0.95–1.04) players based on official HLTV thresholds, we aim to identify the key differences in gaming behavior between elite performing professional players and qualified performing professional players. Alternatively, it can be understood as what metrics distinguish 'professional players with high performance rating' from 'professional players with qualified performance rating'. Additionally, to ensure data robustness and eliminate the influence of insufficient match data, only players with more than 100 maps played in their career were included in the final dataset. This threshold ensures that the game features are based on a meaningful number of professional matches and reflect stable performance patterns. An isolated few players' data appeared as garbled text or system errors in the official system and were excluded during the data preparation process. The data for this study will be provided in the supplementary file.

Ethic consideration

According to the guidelines of the Research Ethics Committee of Universiti Kebangsaan Malaysia (UKM JEP, https://www.ukm.my/jepukm/), this type of study falls outside the scope of ethical review. The research does not involve human trials, medical interventions, any exposure

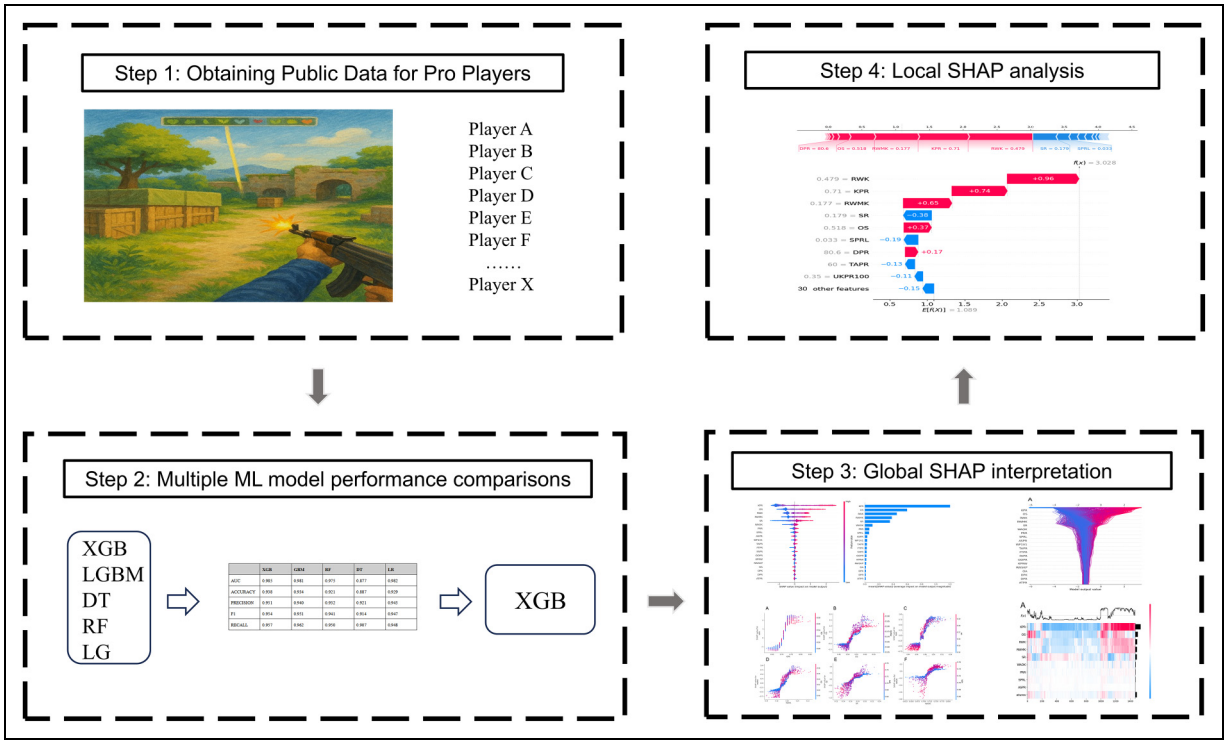


Figure 1. Study design. Step 1: Data sources. Step 2: Multiple modeling and comparison. Step 3: Global SHAP visualization and analysis. Step 4: Local SHAP case diagnosis.

of subjects to physical or psychological risk. This position has been informally confirmed by two members of the Ethics Committee. Instead, it relies entirely on publicly available data from esports platforms. After checking the research process at the Faculty of Education at Universiti Kebangsaan Malaysia, this study could be conducted with the completion of the data statement. All data used in this study were sourced from Half-Life Television (HLTV, https://www.hltv.org/). The data are completely collected for non-commercial academic purposes and used solely for academic research.

Evaluation indicators

The feature variables (X) in this study are derived from HLTV's official Advanced Stats system, specifically under seven tactical skill dimensions: firepower, entrying, trading, clutching, opening, sniping, and utility. Unlike rating, which is a composite performance metric, most of these indicators are original, unpackaged behavioral-level statistics, directly reflecting a player's in-game actions and situational performance.

This study involves 39 indicators under the seven tactical skill dimensions of modeling. Kills per round (KPR), Kills per round win (KPRW), Damage per round (DPR), Damage per round win (DPRW), Rounds with a kill (RWK), Rounds with a multi-kill (RWMK), Pistol round rating (PRR), Saved by teammate per round (SBTPR), Traded deaths per round (TDPR), Traded deaths percentage (TDP), Opening deaths traded percentage (ODTP), Assists per round (ASPR), Support rounds (SR), Saved teammate per round (STPR), Trade kills per round (TKPR), Trade kills percentage (TKP), Assisted kills percentage (AKP), Damage per kill (DPK), Opening kills per round (OKPR), Opening deaths per round (ODPR), Opening attempts (OA), Opening success (OS), Win% after opening kill (WAOK), Attacks per round (ATPR), Clutch points per round (CPPR), Last alive percentage (LAP), 1on1 win percentage (WP1V1), Time alive per round (TAPR), Saves per round loss (SPRL), Sniper kills per round (SKPR), Sniper kills percentage (SKP), Rounds with sniper kills percentage (RWSKP), Sniper multi-kill rounds (SMKR), Sniper opening kills per round (SOKPR), Utility damage per round (UDPR), Utility kills per 100 rounds (UKPR100), Flashes thrown per round (FTPR), Flash assists per round (FAPR), Time opponent flashed per round (TOFPR).

These metrics offer fine-grained assessments of a player's tactical roles and contextual abilities, such as success in entry duels, contribution in clutch rounds, or effective use of utilities. Most of the variables are intuitively interpretable based on their names (e.g., flash assists per round, opening kill success rate). For clarity and transparency, a full list of variable definitions sourced directly from HLTV will be provided in the supplementary materials.

Machine learning and interpretable models

In this study, five industry-established models were selected for comparative analysis: XGBoost, LightGBM, Random Forest, Decision Tree, and Logistic Regression.[28–30] Python was employed in version 3.12.2, with XGBoost version 3.0.2, LightGBM version 4.6.0, and SHAP version 0.48.0.

These models span a methodological range from simple linear classifiers to advanced ensemble learning approaches. All selected algorithms are well-suited for structured tabular data, which aligns with the nature of the performance indicators used in this study.[31] These models have been widely adopted in sports performance analytics due to their balance between predictive accuracy and interpretability.[12,14] To evaluate predictive effectiveness and support explainable modeling, separate model comparisons were conducted on two datasets corresponding to the CSGO (2012–2023) period and CS2 (2023–2025) period. A 5-fold cross-validation procedure was employed to assess performance using metrics such as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). The results confirmed the superior performance of XGBoost across both periods. To further enhance interpretability, SHAP was integrated into the modeling framework to analyze the contribution of individual features within the XGBoost predictions.[32] Due to space limitations, mathematical formulations for LightGBM, Random Forest, Decision Tree, and Logistic Regression are provided in the supplementary materials. This study focuses primarily on the XGBoost model as the main predictive and interpretable modeling approach.

XGBoost. The XGBoost algorithm is based on Gradient Boosted Decision Tree (GBDT), it improves the enhancement algorithm by combining multiple weak classifiers into an optimized classifier.[33] The objective function of XGBoost consists of two parts: a differentiable loss function that measures the model's predictive accuracy, and a regularization term that penalizes model complexity to prevent overfitting. The general form of the objective function is:

$$L(\phi) = \sum_{i=1}^{n} l(\hat{y}_i, y_i) + \sum_{k=1}^{K} \Omega(f_k)$$

Where $l(\hat{y}_i, y_i)$ is a differentiable convex loss function representing the difference between the true label y_i and the predicted value y_i, f_k is the k-th regression tree in the ensemble; $\Omega(f_k)$ is a regularization term defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

T is the number of leaves in the tree, wj is the weight of the j -th leaf, γ is a penalty for using more leaves, and λ controls how much to penalize large leaf weights. This helps XGBoost balance prediction accuracy with model simplicity.

To further optimize training, XGBoost applies second-order Taylor expansion of the loss function and uses gradient-based methods to build trees efficiently. These enhancements lead to superior performance in classification and regression tasks.[33]

SHAP. Machine learning algorithms have demonstrated excellent performance in predicting outcomes in a wide range of domains, but they also face the problem of explainability. We are often unable to understand the decision-making process of machine learning algorithms, often referred to as black-box modeling.[34] To enhance model interpretability, SHAP values were used to explain the output of the XGBoost classifier. SHAP is based on cooperative game theory and assigns each feature an importance value by calculating its marginal contribution across all possible feature combinations.[32,35] We used TreeSHAP for our tree-based primary model (XGBoost). The SHAP value for feature i is defined as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

where N is the set of all features. S is a subset of features excluding i. f(S) is the model prediction using features in set S. SHAP allows us to quantify the positive or negative impact of each feature on the model's output. It also supports intuitive visualizations such as summary plots, dependence plots, and force plots, which help to identify the most influential features and their effect directions. These characteristics make SHAP an effective tool for interpreting complex machine learning models in a transparent and human-understandable manner.[36]

Results

Performance comparison

This study employed five classification performance metrics, AUC, F1 Score, Accuracy, Precision, and Recall, to evaluate the performance and quality of predictive models across two distinct periods: CS:GO (2012–2023) and CS2 (2023–2025). AUC measures the model's ability to distinguish between classes and is widely regarded as the most reliable metric for binary classification tasks. F1 Score provides a harmonic mean of Precision and Recall, balancing false positives and false negatives. Accuracy indicates the overall correctness of predictions. Precision assesses the proportion of true positives among predicted positives, while Recall reflects the model's ability to identify all actual positives.

The results of the five-fold cross-validation comparison among different models are shown in Table 1 and Table 2. In the CSGO phase, XGBoost outperformed all other models across evaluation metrics, with the sole exception of a slightly higher F1-score achieved by Logistic Regression. In the CS2 phase, while LightGBM exhibited a marginally higher recall, XGBoost remained superior in all other metrics. Notably, XGBoost consistently achieved the highest AUC values across both phases. Given that this study focuses on a binary classification task, AUC is considered the most critical evaluation metric.

Table 1. Comparison of the performance of four machine learning models during the CS:GO period (2012–2023).

	XGB	LGBM	RF	DT	LR
AUC	0.990	0.987	0.983	0.888	0.954
ACCURACY	0.952	0.944	0.934	0.905	0.896
PRECISION	0.925	0.923	0.902	0.849	0.858
F1	0.923	0.907	0.892	0.846	0.825
RECALL	0.922	0.893	0.883	0.845	0.797

Table 2. Comparison of the performance of four machine learning models during the CS2 period (2023–2025).

	XGB	LGBM	RF	DT	LR
AUC	0.985	0.981	0.975	0.877	0.982
ACCURACY	0.938	0.934	0.921	0.887	0.929
PRECISION	0.951	0.940	0.932	0.921	0.945
F1	0.954	0.951	0.941	0.914	0.947
RECALL	0.957	0.962	0.950	0.907	0.948

Overall, XGBoost demonstrated superior predictive performance in both periods. Considering its strong compatibility with SHAP-based interpretability analysis, XGBoost was selected as the primary model for this study.

Feature correlation and descriptive statistics

The descriptive statistics of the indicators for both CSGO and CS2 periods are shown in Table 3. Details of the metric distribution can be seen in the box plot visualization in the supplementary materials. Due to the fact that many metrics share numerous related events within the same capability dimension, the numerator or denominator of their calculations may overlap, resulting in significant strong correlations. For instance, metrics related to firepower such as kills and damage, as well as sniper metrics tied to sniping ability. These correlations align with dependencies arising from roles or tactics, and heatmaps clearly reveal that highly correlated metrics cluster within their respective capability dimensions.

While certain gaming behavior indicators used in this study exhibit significant correlations as shown in Figures 2 and 3, we chose not to perform feature elimination based on these correlations for two key reasons. First, all selected indicators originate from official assessment system established by authoritative institutions. Removing any single indicator could disrupt the internal consistency and conceptual structure of the evaluation system. Second, the objective of this research is to investigate the interpretability and predictive dynamics of the model within the context of the full original indicator set, rather than to optimize performance by selectively reducing input features.[37] This approach ensures that the interpretability results remain comprehensive, coherent, and aligned with official evaluation standards,[38] which is particularly critical for the SHAP-based analysis of individual feature contributions in later sections. Moreover, this aligns with the objective of this study, which is to fully interpret advanced statistical indicators rather than prune features based on correlation. Therefore, Figure 2 is presented only for descriptive purposes, offering a visual reference for understanding the interrelationships among the indicators.

Global model interpretation with SHAP

The distribution of SHAP values generated by the XGBoost model across two classification periods as shown in Figures 4 and 5, provides insight into how various features influence the prediction outcomes. On the X-axis, the SHAP value quantifies the strength and direction of each feature's contribution to the model output. On the Y-axis, features are ranked based on their overall impact on predictions. A color gradient is applied to reflect feature value magnitudes, cool colors (blue) correspond to lower values, while warm colors (red) represent higher values. This form of visualization not only highlights which features are most influential but also reveals how different value ranges contribute to prediction shifts, offering a nuanced understanding of the model's decision-making process.[14]

During the CSGO period (2012–2023), the top ten metrics that have the greatest impact on the model are Kills Per Round (KPR), Opening Success (OS), Rounds With a Kill (RWK), Rounds With a Multi-Kill(RWMK), Support Rounds (SR), Win% After Opening Kill (WAOK), Pistol Round Rating (PRR), Saves Per Round Loss (SPRL), Assists Per Round (ASPR), 1on1 Win Percentage (WP1V1).

During the CS2 period (2023–2025), the top ten metrics that have the greatest impact on the model are Rounds With a Kill (RWK), Kills Per Round (KPR), Rounds With a Multi-Kill (RWMK), Opening Success (OS), Support Rounds (SR), Damage per kill (DPR), Saves Per Round Loss (SPRL), Traded deaths percentage (TDP), Pistol Round Rating (PRR), Time alive per round (TAPR).

Figure 6 illustrates the SHAP decision plot for the CSGO (2012–2023) and CS2 (2023–2025) periods, which are used to illustrate how the model integrates the contributions of individual features across multiple samples to generate the final predictions. Each curve in the graph represents a sample whose path starts from the model's baseline prediction and is progressively influenced by different features, eventually reaching the actual prediction for that sample. The horizontal offset indicates the increasing or decreasing effect of a feature on the prediction result, while the color gradient reflects the high or low value of the feature, thus providing an intuitive perception of the feature structure of the sample. It also reveals the differences in the model's prediction mechanisms across samples, as well as the consistency and influence distribution of multiple key features in the overall prediction decision.

The SHAP dependence plot helps to determine whether the model presents a linear or nonlinear relationship to the

Table 3. Descriptive statistics of indicators during the CSGO period (2012–2023) and CS2 period (2023–2025).

	CSGO period (n = 1468)			CS2 period (n = 880)		
		95% Confidence interval			95% Confidence interval	
Indicators	Mean (S.D)	Lower	Upper	Mean (S.D)	Lower	Upper
Firepower						
KPR	0.699 (0.032)	0.697	0.700	0.679 (0.045)	0.676	0.682
KPRW	0.970 (0.052)	0.967	0.972	0.934 (0.065)	0.930	0.939
DPR	75.788 (3.719)	75.598	75.978	74.358 (4.610)	74.054	74.663
DPRW	93.402 (13.113)	92.731	94.073	96.689 (6.780)	96.241	97.137
RWK	0.472 (0.016)	0.471	0.472	0.464 (0.022)	0.463	0.466
RWMK	0.171 (0.011)	0.170	0.172	0.163 (0.015)	0.162	0.164
PRR	1.081 (0.091)	1.076	1.085	1.075 (0.105)	1.068	1.082
Entrying						
SBTPR	0.095 (0.011)	0.095	0.096	0.104 (0.013)	0.103	0.105
TDPR	0.124 (0.022)	0.123	0.125	0.134 (0.018)	0.133	0.136
TDP	0.186 (0.030)	0.184	0.187	0.186 (0.030)	0.184	0.187
ODTP	0.228 (0.047)	0.226	0.231	0.257 (0.038)	0.254	0.259
ASPR	0.126 (0.019)	0.125	0.127	0.185 (0.031)	0.183	0.187
SR	0.168 (0.017)	0.167	0.169	0.201 (0.020)	0.199	0.202
Trading						
STPR	0.100 (0.013)	0.099	0.101	0.107 (0.014)	0.106	0.108
TKPR	0.131 (0.022)	0.130	0.132	0.138 (0.015)	0.137	0.139
TKP	0.188 (0.031)	0.187	0.190	0.203 (0.020)	0.202	0.204
AKP	0.184 (0.014)	0.183	0.185	0.269 (0.018)	0.267	0.270
DPK	102.800 (13.526)	102.108	103.492	109.463 (4.469)	109.168	109.758
Opening						
OKPR	0.104 (0.017)	0.103	0.105	0.101 (0.019)	0.1030	0.103
ODPR	0.097 (0.018)	0.096	0.098	0.097 (0.023)	0.096	0.099
OA	0.201 (0.031)	0.199	0.203	0.199 (0.037)	0.197	0.202
OS	0.517 (0.040)	0.515	0.519	0.511 (0.051)	0.508	0.514
WAOK	0.725 (0.024)	0.724	0.726	0.722 (0.029)	0.720	0.724
ATPR	2.596 (0.387)	2.576	2.615	2.639 (0.359)	2.615	2.663
Clutching						
CPPR	0.023 (0.005)	0.022	0.023	0.022 (0.006)	0.022	0.023
LAP	0.103 (0.024)	0.101	0.105	0.105(0.025)	0.103	0.106
WP1V1	0.605 (0.054)	0.602	0.608	0.608 (0.069)	0.604	0.613
TAPR	60.497 (10.024)	59.984	61.010	65.227 (4.355)	64.939	65.515
SPRL	0.060 (0.022)	0.059	0.061	0.075 (0.025)	0.073	0.077
Sniping						
SKPR	0.108 (0.139)	0.101	0.115	0.072 (0.125)	0.063	0.080
SKP	0.153 (0.195)	0.143	0.163	0.103 (0.177)	0.091	0.115
RWSKP	0.082 (0.105)	0.077	0.087	0.056 (0.096)	0.050	0.062
SMKR	0.028 (0.037)	0.026	0.030	0.018 (0.032)	0.016	0.020
SOKPR	0.024 (0.032)	0.022	0.026	0.016 (0.029)	0.014	0.018
Utility						
UDPR	3.946 (1.202)	3.855	4.008	4.494 (1.225)	4.413	4.575
UKPR100	0.830 (0.268)	0.817	0.844	0.832 (0.243)	0.816	0.848
FTPR	0.553 (0.148)	0.546	0.561	0.548 (0.152)	0.538	0.558
FAPR	0.035 (0.018)	0.035	0.036	0.042 (0.020)	0.041	0.044
TOFPR	2.052 (0.459)	2.029	2.076	2.271 (0.198)	2.258	2.284

feature. Moreover, this helps us to understand the trend of the marginal contribution of the primary feature to the model, as well as to find out whether there is an interaction effect between the two variables through the color-represented secondary feature.[39] In interpretable modeling, it is more reasonable to understand the mechanism behind the model rather than just stopping at feature importance ranking. For this purpose, we identified SHAP dependency plots for the first six continuous variables in the XGBoost model for both periods.

Figure 7 demonstrates a positive correlation between SHAP values and top six features during the CS:GO period, including Kills Per Round (KPR), Opening Success (OS), Rounds With a Kill (RWK), Rounds With a Multi-Kill

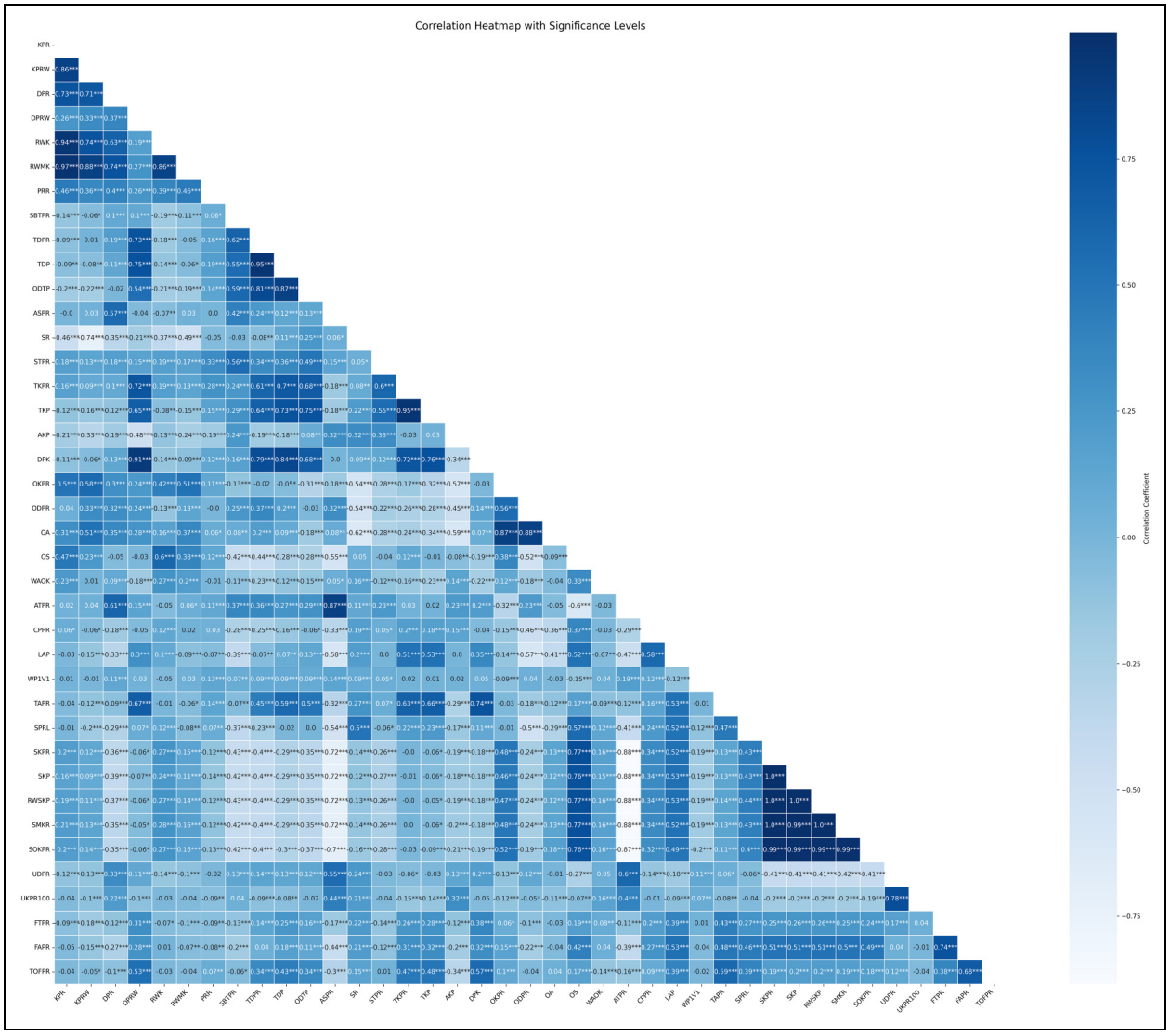


Figure 2. Heatmap of correlation on game behavior metrics. CSGO period 2012–2023; The asterisks reflect the significance levels of the correlation coefficients: no asterisk denotes $p > 0.05$, one asterisk denotes $0.01 < p < 0.05$, two asterisks denote $0.001 < p < 0.01$, and three asterisks denote $p < 0.001$.

(RWMK), Support Rounds (SR), and Win% After Opening Kill (WAOK). The observed thresholds for these features are approximately 0.71, 0.53, 0.47, 0.17, 0.17, and 0.725.

Figure 8 shows a similar pattern during the CS2 period, where Rounds With a Kill (RWK), Kills Per Round (KPR), Rounds With a Multi-Kill (RWMK), Opening Success (OS), Support Rounds (SR), and Damage Per Round (DPR) are all positively associated with SHAP values. The corresponding thresholds are approximately 0.46, 0.67, 0.17, 0.48, 0.21, and 73.

The SHAP heatmap is used to globally interpret the model structure and determine whether key features have a dominant influence on the prediction results for most samples.[40] The X-axis represents different samples, and the Y-axis represents the top ten features most important to the model. The color represents the degree of influence of each feature on the prediction for each sample, and f(x) represents the final prediction value of the model for each sample. Figure 9 illustrates the strength and direction of the influence of each feature on the model output across multiple samples. This helps identify the features that play a critical role in most samples and whether the influence mechanisms are consistent across samples.

Case diagnosis and analysis

To further examine the model's prediction mechanism at the individual level, a representative player from the CS2 period was selected for case diagnosis analysis using the force and waterfall plots as shown in Figure 10. SHAP

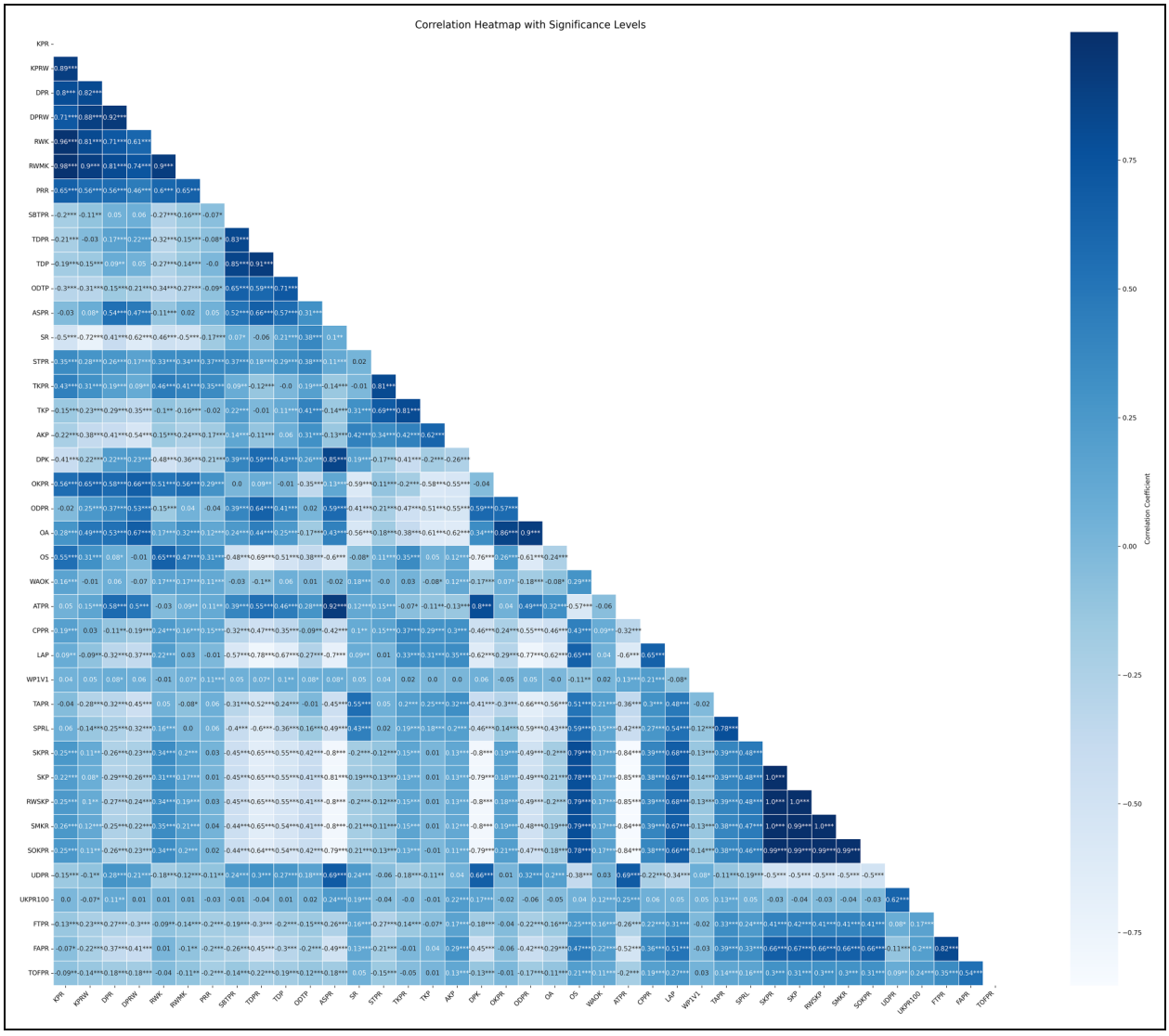


Figure 3. Heatmap of correlation on game behavior metrics. CS2 period 2023–2025; The asterisks reflect the significance levels of the correlation coefficients: no asterisk denotes $p > 0.05$, one asterisk denotes $0.01 < p < 0.05$, two asterisks denote $0.001 < p < 0.01$, and three asterisks denote $p < 0.001$.

values indicate how each feature pushes the model's output toward or away from the 'good' class (Rating ≥ 1.05) for this player. This approach clearly illustrates the positive and negative contributions of each feature, showing how the model integrates multiple performance metrics to make classification decisions.[12]

As shown in the force plot, the model's baseline prediction of 1.089 is driven upward to 3.028 by key offensive features such as RWK (0.479), KPR (0.71), RWMK (0.177), and OS (0.518). In contrast, SR (0.179) and SPRL (0.033) slightly reduce the prediction score. The waterfall plot quantifies these effects, RWK (+0.96) and KPR (+0.74) are the strongest positive contributors, followed by RWMK (+0.65) and OS (+0.37), while SR (−0.38) and SPRL (−0.19) have limited negative impact. It can be seen that the performance of this player's firepower and opening abilities are the key factors for the model to make high prediction judgments, while the other metrics have a weaker influence.

This case diagnosis analysis highlights the model's strong interpretability and provides insight into how it evaluates individual player performance, especially emphasizing the importance of metrics in high-performance classification.

Discussion

The main purpose of this study is to explore and distinguish the evolving trends in key performance indicators that define excellent player performance across different developmental periods of the Counter-Strike (CS). To date, this is the first longitudinal study analyzing professional CS

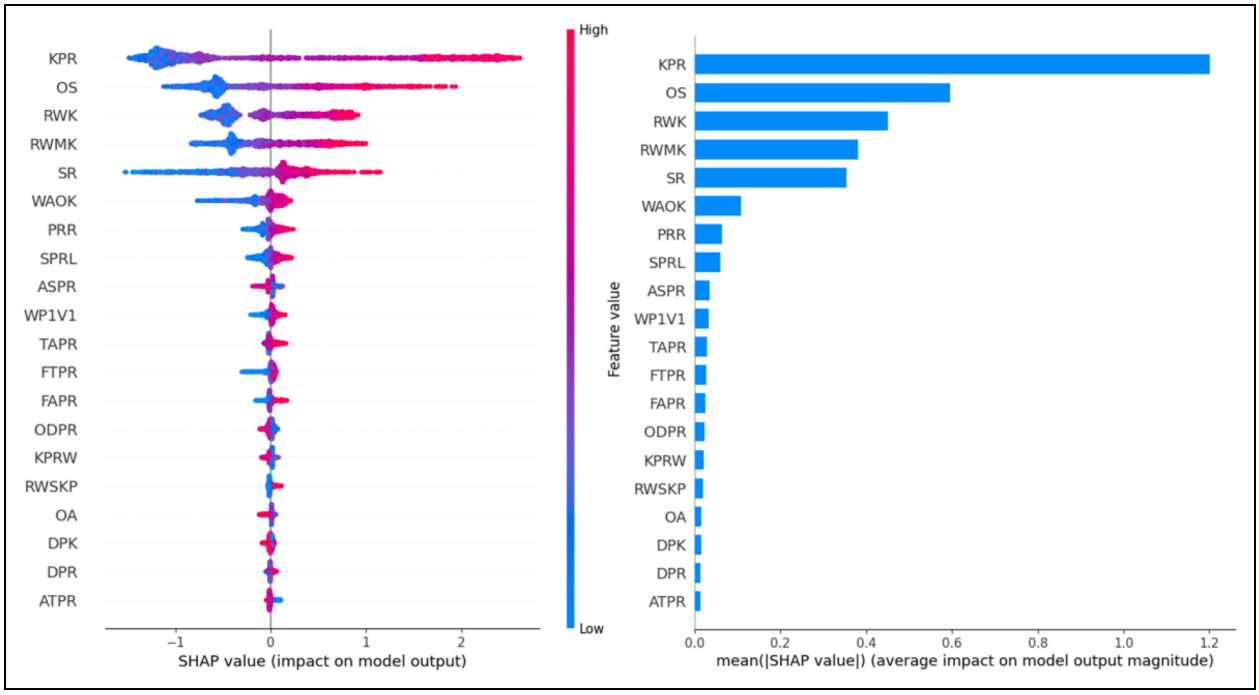


Figure 4. SHAP summary plot among CSGO period (2012–2023).

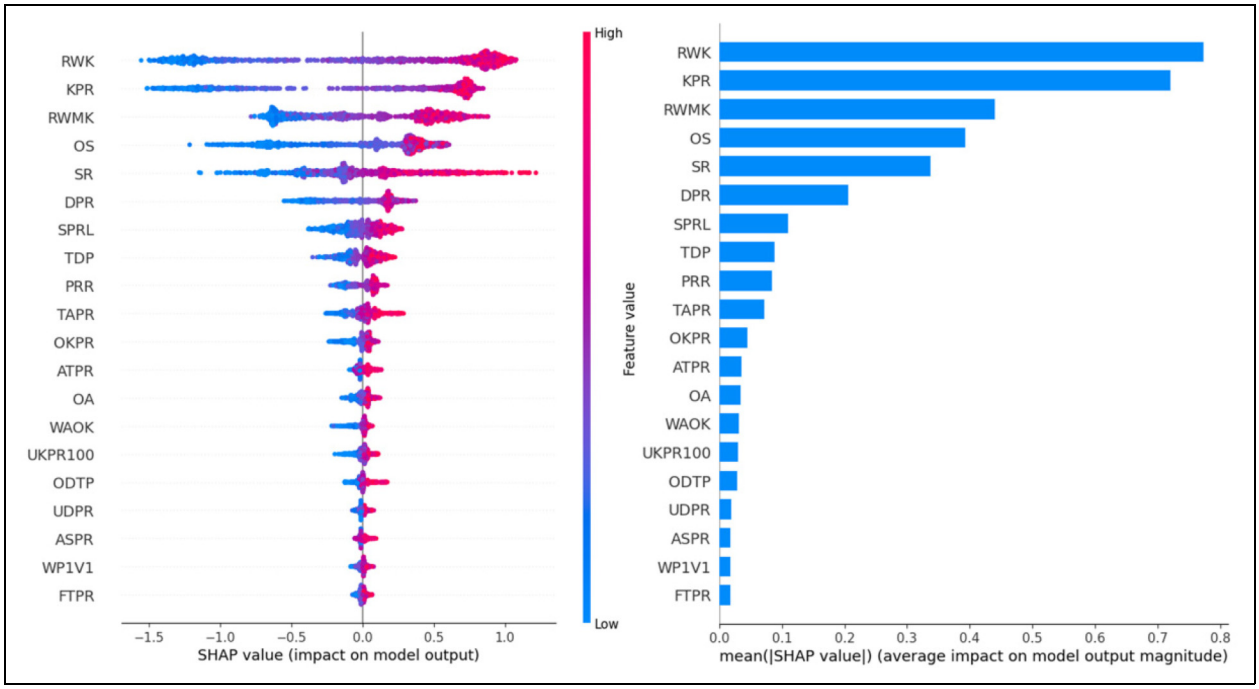


Figure 5. SHAP summary plot among CS2 period (2023–2025).

players' performance across distinct eras. By employing a multi-model machine learning framework combined with SHAP-based interpretability analysis, this research offers a novel perspective to performance evaluation in the field of esports analytics.

The SHAP importance analysis reveals that although the competitive esports environment has transitioned from CSGO (2012–2023) to CS2 (2023–2025), the behavioral patterns that constitute high-level player performance exhibit both stability and structural change across the two

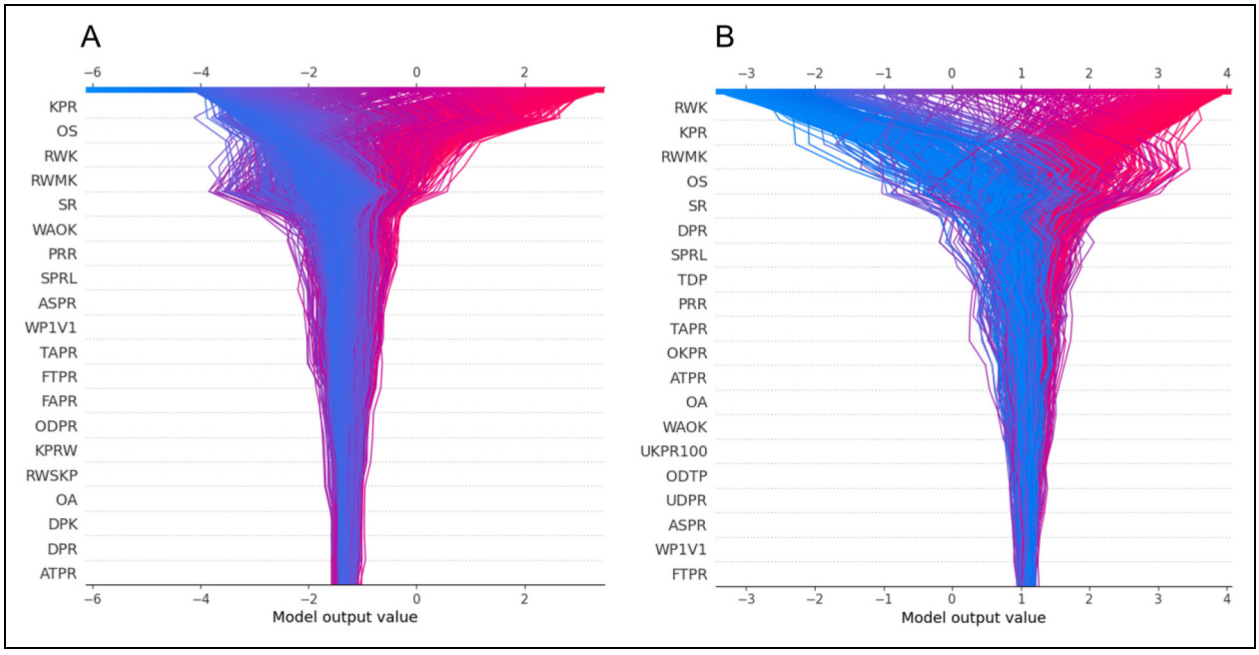


Figure 6. SHAP summary plot among CSGO and CS2 period. A represents the CSGO period (2012–2023). B represents the CS2 period (2023–2025).

periods. Specifically, Kills Per Round (KPR), Opening Success (OS), Rounds With a Kill (RWK), Rounds With a Multi-Kill (RWMK), and Support Rounds (SR) consistently rank among the top five most important indicators in both eras. This suggests that in a competitive context, high offensive output, early-round impact, and sustained threat remain core behavioral traits in the FPS game,[5,24] A key finding is that during the CSGO phase, KPR significantly outperformed other metrics in importance, while in the CS2 phase, RWK became the most important indicator, with both RWK and KPR showing much higher SHAP values than other features. This shift reflects not only changes in the rating system structure but also an evolving philosophy in competitive standards and player evaluation. In the CSGO phase, HLTV Rating 1.0 placed heavy emphasis on kill count and multi-kill actions, assigning greater weight to raw fragging while comparatively undervaluing survival, assists, and team participation.[24] The dominance of KPR during this period highlights the emphasis on high-frequency burst output as the core criterion for elite performance. Players who could consistently produce kills were rated highly, encouraging a strategy that prioritized individual fragging ability. However, in the CS2 phase, RWK surpassed KPR as the primary factor influencing the rating, a change closely tied to the structural refinement in HLTV Rating 2.0. This newer system incorporates a broader set of metrics (such as KAST, ADR, and Impact), placing greater importance on round-by-round involvement and consistency.[41] RWK measures the percentage of rounds in which a player records at least one kill, focusing not on kill volume but on the frequency of meaningful contributions. This reflects a shift in evaluation from burst performance to consistent impact. The rise of RWK also indicates a growing emphasis on tactical execution, team coordination, and role adaptability in the current CS2 competitions.[42] In this version, players who can deliver stable output across most rounds, even without frequent multi-kills, are valued more highly than those with occasional standout moments. It's also a response to the progression of the game's development in which professional players have increasingly favored integrated performance, team synergy, and tactical consistency.

It is noteworthy that Saves Per Round Loss (SPRL) and Pistol Round Rating (PRR) consistently appear among the top ten SHAP importance metrics across both periods, indicating the sustained and cross-version relevance of economic impact and early round performance in player evaluation. PRR measures a player's performance in pistol rounds at the beginning of each half rounds that carry the greatest economic influence within a match.[43] The consistently high importance of PRR underscores the strategic weight of pistol rounds in the match structure.[43] Previous studies have clearly shown that winning the pistol round in the first and second halves to gain an economic advantage is an important dimension for evaluating player performance.[24] SPRL, on the other hand, reflects a player's survival rate during lost rounds, often indicating their ability to preserve weapons or utility for the following round, thereby saving economy or maintaining firepower. In high-level competitive esports matches, economic management is a crucial component of match control.[44] By increasing survival on weak rounds, the team can regain some gun

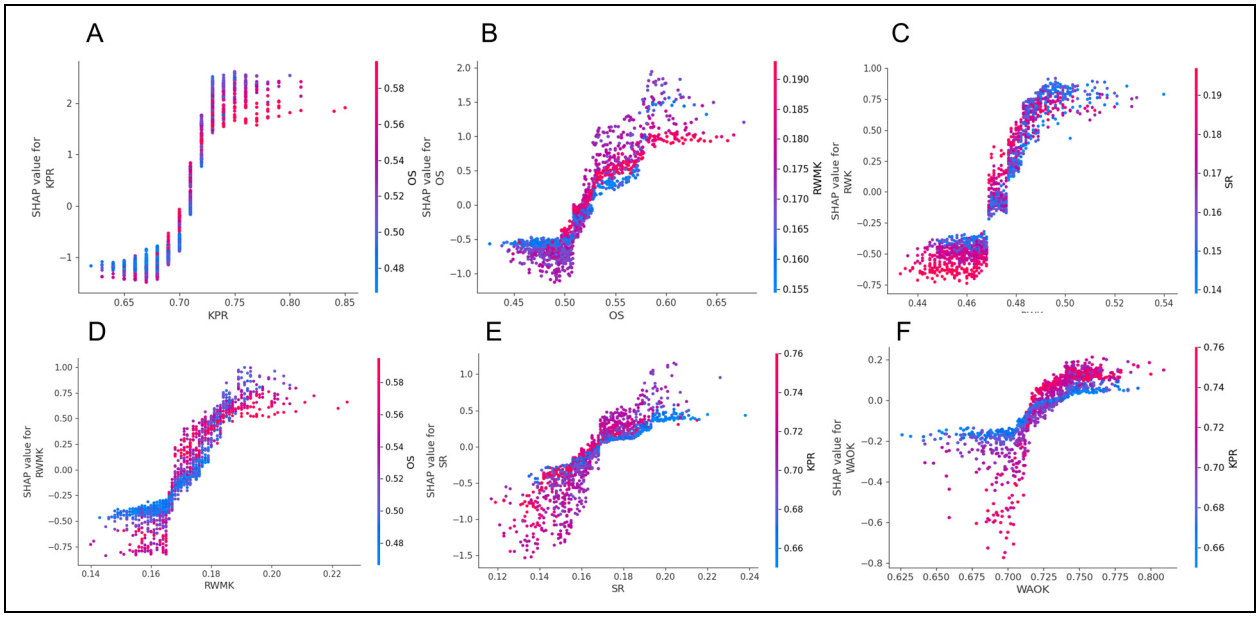


Figure 7. SHAP dependency plot of the top six variables in the XGBoost model for the CSGO period.

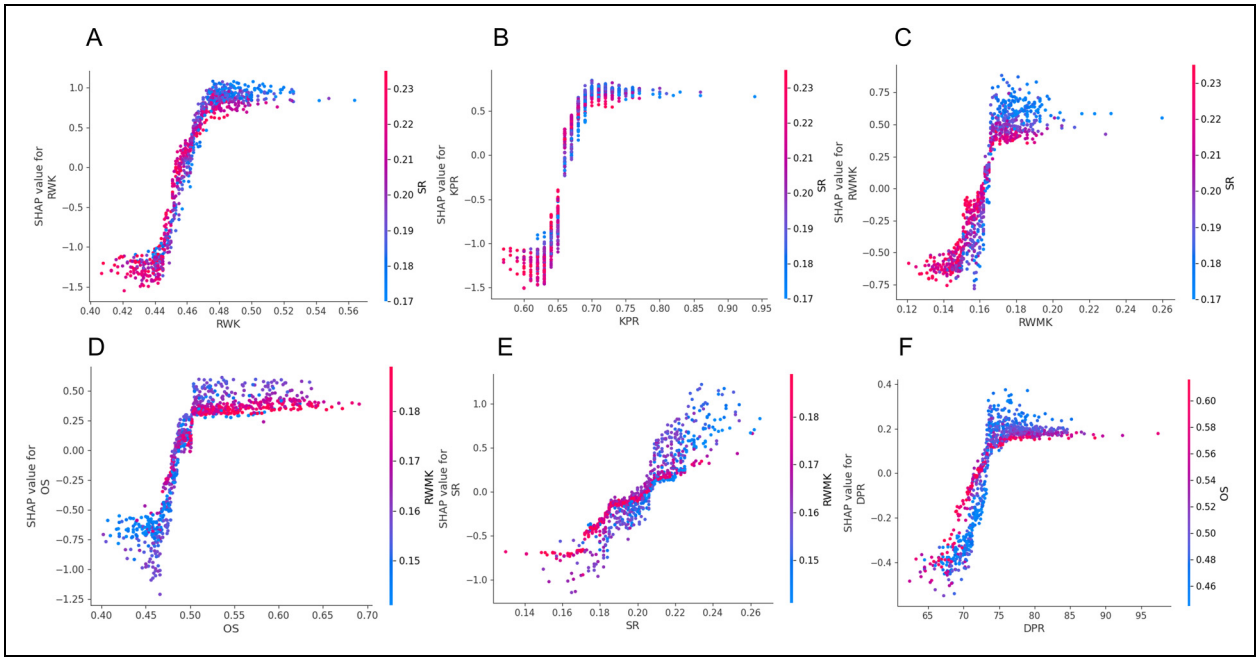


Figure 8. SHAP dependency plot of the top six variables in the XGBoost model for the CS2 period.

disadvantage and economic balance through the act of gun and utilities preservation.[45–47] In addition, upon entering the CS2 phase, the rising importance of several new features reflects an evolution in tactical philosophy and team structure. The rise in importance of non-firepower metrics such as Time Alive Per Round (TAPR), Utility Damage Per Round (UDPR), and Flashes Thrown Per Round (FTPR) not only reflects teamwork and tactical discipline in esports, but also highlights survival and utilities capabilities.[48] There have been studies examining the use of utilities to predict dominance in obtaining a victory in CS2 competitions,[15] demonstrating the importance of utilities' use in pro-level matches. The effective utility use and timing, such as smoke grenades, molotovs, and flashbangs is closely linked to tactical coordination and team strategy among Pro-players.[8] The increased importance of Time

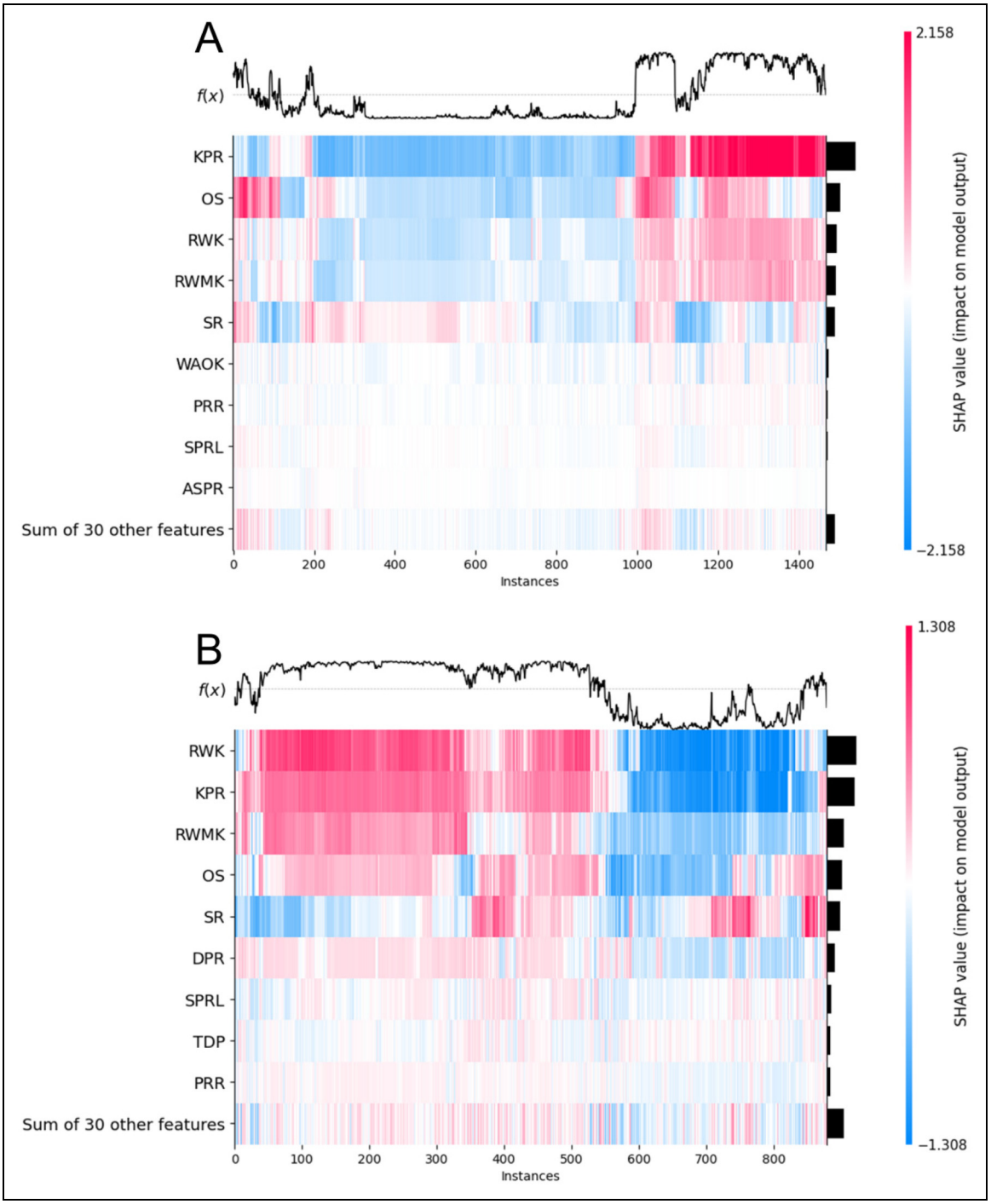


Figure 9. SHAP heatmap plot in the XGBoost model. A represent CSGO period (2012–2023), B represents the CS2 period (2023–2025).

Alive per Round (TAPR) and Saves Per Round Loss (SPRL) suggest that the current version places greater emphasis on survivability. The elevated status of the survival indicator reaffirms the importance of the CS2 period to the role of survivability in rating player performance.[24]

The SHAP dependence results for the top six key performance metrics across both periods demonstrate a consistently positive correlation between feature and SHAP values. This indicates that improvements in these indicators reliably increase a player's likelihood of being classified as an elite performer by the model. Moreover, the presence of positive trends and threshold effects provides specific interpretability of indicators.[49] By identifying threshold and turning point, we can clearly determine when

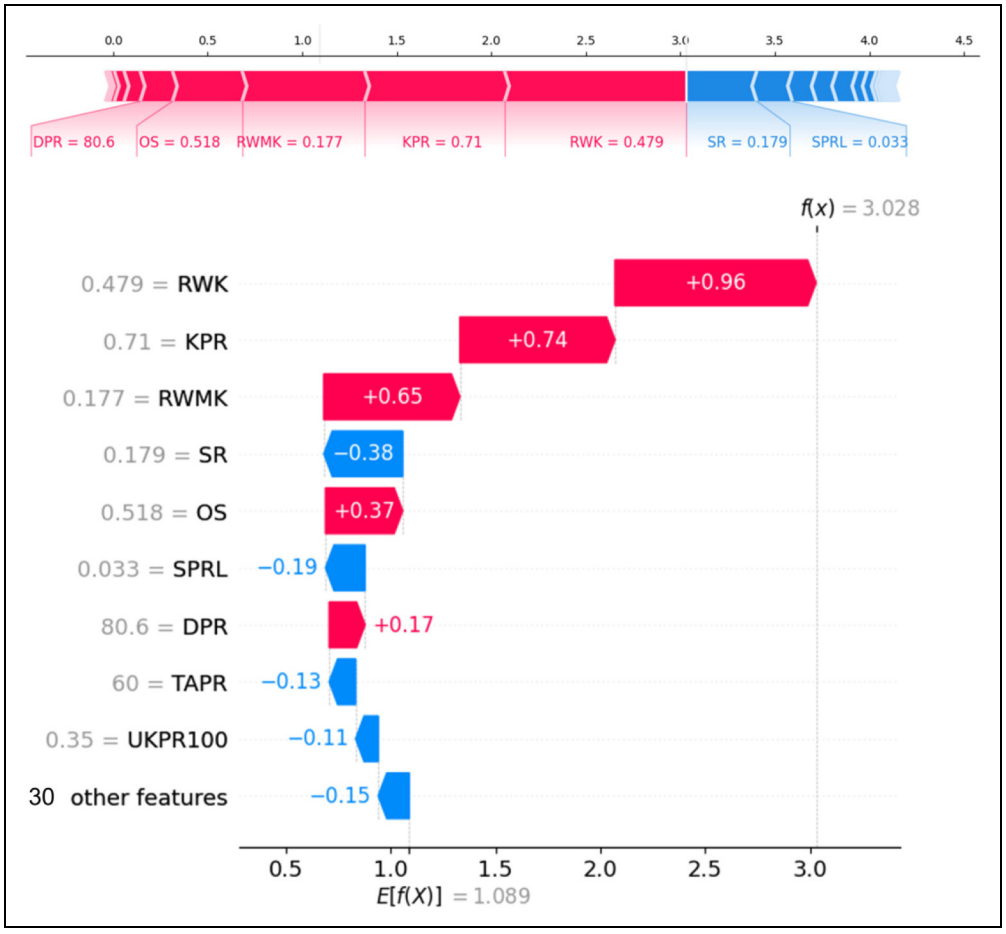


Figure 10. A case diagnostic example of a CS2 period player.

a given metric begins to exert a significant influence on the classification of elite performance. The interactions between metrics not only offer macro-level insights into players' tactical performance, but also provide micro-level explanations of how specific combinations of actions amplify performance impact.[36] These findings reveal the nonlinear dynamics in technical performance metrics. Additionally, the individual diagnostic cases in this study demonstrate how SHAP-based explainability methods can yield actionable micro-level insights tailored to specific players. This approach complements global feature interpretations and illustrates how case diagnosis can directly support talent identification frameworks in competitive sports.[12] Rather than relying solely on overall ratings to assess player performance, this approach enables precise identification of strengths and weaknesses in specific skill dimensions,[12] facilitating targeted evaluation, training optimization, and talent scouting based on case diagnosis. Unlike traditional sports such as football or basketball, where identification or sports analysis systems, biometric tracking and performance benchmarks are already standardized.[50–52] Esports still lacks structured and data-driven mechanisms for player assessment, with selection processes often remaining opaque and ill-defined.[53,54]

Our research integrates machine learning models with an interpretable framework to generate multidimensional, explainable performance profiles for each esports player. This approach has already been validated in traditional sports performance analytics, such as football, basketball, and table tennis.[12,14,36] Coaches and team managers can use these profiles to design personalized training roadmaps aligned with the skill emphases and benchmarks of elite professional esports players, helping to cultivate underdeveloped dimensions of performance. Moreover, this framework is domain-adaptive and can be applied to other esports programs with similarly structured player performance data. As such, it provides a valuable supplement to scientific esports talent development. This is essential for continued professionalization, talent identification and scientific training in esports.[55]

This study presents several novel contributions to the evolving field of esports performance analysis. Firstly, by employing machine learning models and interpretable SHAP analysis, we not only identified the key performance

metrics across two major game eras but also uncovered their nonlinear threshold effects and interaction. This longitudinal analytical perspective enables a deeper understanding of how player metrics evolve in response to game updates, development of the game style, and tactical system evolution, thereby offering actionable insights for coaches, analysts, and team managers in tactical evaluation and talent identification. Compared to traditional statistical analysis, SHAP provides actionable player-level explanations beyond global importance. This is because the research paradigm aims to perform predictive diagnostics within multivariate, correlated feature spaces, which traditional statistics cannot encompass. However, this study also has certain limitations. Primarily, the modeling focused on distinguishing between professional players rated as good and average level, which may restrict the generalizability of the findings to a broader player base. We also recognize that current research focuses on individual and global levels, while future work should attempt to explore meso-level analyses within entire teams. Additionally, the CS2 phase presents a temporal limitation. Our analysis of the CS2 phase may be influenced by ongoing changes in game mechanics, and the importance of certain key metrics could shift with future updates or tactical evolutions. Therefore, the findings related to the CS2 era should be verified over a longer observation period to ensure their explanatory stability.

Conclusion

Performance analysis of elite athletes is no longer exclusive to traditional sports. With the growing professionalization of esports, analyzing the performance of professional esports players, particularly in competitive programs like Counter-Strike (CS) has become increasingly essential. Therefore, we analyze the performance of professional players in two periods of the Counter-Strike program from 2012–2025. This study adopted a SHAP-based interpretability framework to analyze the outputs of machine learning models and diagnose player performance across both eras of professional Counter-Strike players. Our primary findings indicate that among five classic machine learning models, XGBoost demonstrated the best performance of five-fold cross-validation. Key performance indicators such as Kills Per Round (KPR), Opening Success (OS), Rounds With a Kill (RWK), Rounds With a Multi-Kill (RWMK), Support Rounds (SR), Pistol Round Rating (PRR), and Saves Per Round Loss (SPRL) consistently ranked among the top ten most important features across both periods. However, there is a change in the status and ranking of the importance of the game features. SHAP dependency plots revealed nonlinear, positively correlated relationships between the top six features and SHAP values in both eras. In addition, our case diagnostics offer practical insights for precisely identifying individual players' strengths and weaknesses across specific features' dimensions. These findings and methodological innovations applied in esports provide new practical tools, diagnostic approaches, and algorithmic benchmarks for performance analysis of professional CS players. It also provides a novel and revealing scientific perspective on talent assessment, identification, and training in the evolving field of esports.

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Ethical Considerations

This article does not contain any studies with human or animal participants.

Consent to Participate

Not applicable.

Consent for Publication

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Data Availability Statement

The dataset used in this study will be provided via a GITHUB link in the Supplementary Materials. The core code is available via a GITHUB link in the Supplementary Materials.

Supplemental material

Supplemental material for this article is available online.

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